

Mid-Infrared Imaging and Machine Learning Approaches for Non-Destructive and Efficient Material Detection of Textile Fabrics

TXD 401: Major Project Part I

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Certificate

This is to certify that this project report entitled “**Mid-Infrared Imaging and Machine Learning Approaches for Non-Destructive and Efficient Material Detection of Textile Fabrics**”, being submitted by **Lakshya Kumar** (Entry No. 2022TT12183) and **Keshav Chachan** (Entry No. 2022TT12148), to the Department of Textile and Fibre Engineering, Indian Institute of Technology Delhi, New Delhi, India, for completion of Major Project Part I (TXD401) is a bona fide record of research work carried out by them under my guidance and supervision.

To the best of my knowledge, the results embodied in this project have not been submitted in part or full to other University or Institute.

Signature of Supervisor

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Abstract

Managing textile waste is a big problem in India. Every year, millions of tons of it are made, but very little of it is recycled well. The manual sorting process is a major bottleneck because it takes a lot of time and is easy to make mistakes. This project suggests a way to use Mid-Infrared (MIR) imaging and machine learning to find and sort textile materials (Cotton, Polyester, Nylon) without damaging them.

The first tests were done with passive MIR imaging that used a YOLO11x-cls architecture. The model performed well on controlled validation sets, but it wasn't very robust in different environmental conditions because of thermal noise. To overcome these constraints, the research methodology was deliberately shifted to Terahertz (THz) spectroscopy. Initial findings suggest that the Terahertz range provides enhanced dye-independence and noise resilience. This report talks about how the MIR framework was made, how it was tested, how optical limits were found, and how it was successfully moved to Terahertz characterization for strong textile sorting.

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Chapter-1

Introduction

Introduction

Textile waste management continues to be a major environmental challenge in India, with nearly 7.8 million tons generated annually. A significant portion of this waste—approximately 34%—is landfilled, while only a very small fraction (<1%) undergoes chemical or biological recycling. One of the biggest obstacles preventing large-scale recycling is **material identification**.

Most of the time, India’s textile sorting industry still relies on people (90–95%), which makes processing slow, requires a lot of workers, and makes sorting accuracy inconsistent. To make sure that high-quality recycling streams are possible, we need automated, non-destructive, and scalable identification technologies.

Most industrial solutions today use **Near-Infrared (NIR) spectroscopy**. This method is quick, but it often has trouble telling chemically similar fibers apart and is easily affected by dyes, finishes, and the properties of the fabric surface. Recent studies indicate that **Mid-Infrared (MIR) spectroscopy**, which detects fundamental vibrational modes, offers significantly enhanced chemical specificity for fiber identification. **Tera-hertz (THz) spectroscopy** is also becoming a useful tool for studying bulk structural features. It can go deeper and cause less dye interference.

The goal of this project is to improve automated textile identification by creating a pipeline that combines **MIR imaging, THz spectroscopy insights from the literature, and machine learning models**. The goal is to make textile recycling more accurate, reliable, and useful in the real world.

Chapter-2

Literature Review

Literature Review

Textile fibers are common in everyday life, and their classification and identification are essential in textile recycling, archeology, cultural heritage protection, and public safety. Traditional textile fibres detection methods include visual inspection, microscopic observation, and combustion. With the continuous improvement and updating of detection technology, the current detection methods for textile fibres are chromatography and spectral methods. Chromatography damages the textile fibres and cannot maintain samples integrity [6]. Traditional methods are time-consuming, labor-intensive, and damaging to the samples, hence their use in extensive or delicate situations are limited [6]. On the other hand, spectroscopic methods like Fourier Transform Infrared (FTIR) spectroscopy [2], Near-Infrared (NIR) spectroscopy [3][4], and hyperspectral imaging combined with deep learning [5][6] have become swift, reliable, and non-destructive ways to classify and sort fibers. Recent progress shows that spectroscopy, when used with modern machine learning techniques, has a lot of potential to get around the problems we have with identifying textile materials and make recovery more efficient.

2.1 AN OVERVIEW OF BASICS OF VIBRATIONAL SPECTROSCOPY

Textile fibers' molecular bonds vibrate in certain ways (stretching, bending, twisting) at certain energy levels. When molecules take in electromagnetic radiation at frequencies that match these vibrational energies, they move to higher vibrational states. This is what infrared (IR) spectroscopy is based on.

2.1.1 Mid-Infrared (MIR) Spectroscopy ($\approx 4000 - 400 \text{ cm}^{-1}$)

Principle: MIR spectroscopy only looks at fundamental vibrations, like the $C = O$ stretch, $N - H$ bend, and $C - O$ stretch. These are the most likely and energetically favored transitions from a molecule's ground vibrational state ($v = 0$) to its first excited state ($v = 1$).

Band Characteristics: These transitions that are "allowed" by quantum mechanics create strong, sharp, and specific absorption bands, which make a unique chemical "fingerprint".

Chemical Specificity: The basics of MIR can tell you with a lot of certainty which functional groups are present, such as Amide I/II in proteins and polyamides, Ester $C = O$ in polyesters, $C-O$ stretches in cellulose, and Nitrile $C \equiv N$ in acrylics.

Direct Insight: The direct observation of these fundamental vibrations yields highly interpretable chemical information, rendering MIR inherently more effective for specific chemical identification than NIR.

2.1.2 Near-Infrared (NIR) Spectroscopy ($\approx 780\text{--}2500\text{ nm}$)

Principle: NIR spectroscopy mainly picks up overtone and combination bands, which are higher-energy, less likely vibrational transitions that mostly involve $X\text{--}H$ bonds ($C\text{--}H$, $O\text{--}H$, $N\text{--}H$).

Band Characteristics: These less likely transitions create broad, weak, and very overlapping absorption bands. This lower intensity and wider range make it much harder to assign peaks to distinct functional groups based on visual inspection alone.

Complexity: The resulting spectrum is a complicated, twisted profile that usually needs advanced chemometric models to understand.

The main difference between MIR and NIR is: MIR directly measures fundamental vibrations, which gives it stronger, more specific signals. NIR depends on overtone and combination bands that are weaker and less specific. NIR is faster, can go deeper, and is easier to carry, but MIR is always better at identifying textiles [3].

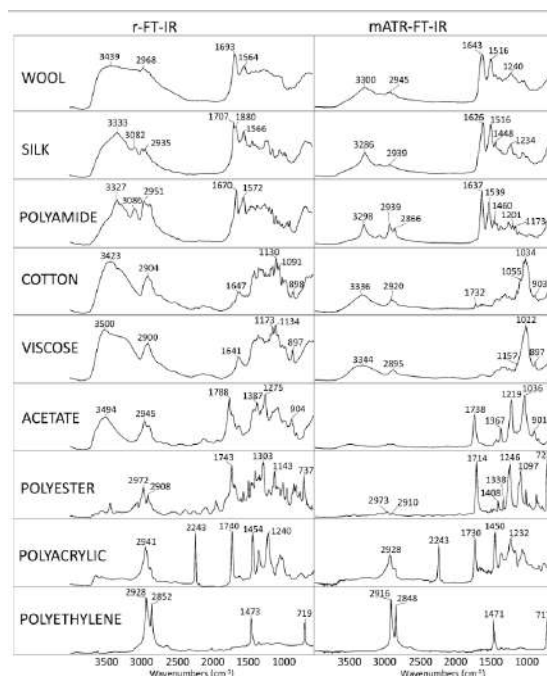


Figure 1: r-FT-IR and mATR-FT-IR spectra of most common textile fibers. Spectra in figure represent normalized and averaged results of each class. The vertical axis represents absorbance. As natural cellulose-based fibers have very similar spectra, only cotton is presented. Spectra of other natural fibers like linen, jute and sisal are shown in Additional file . Source: [2]

Table 1: Description of samples and datasets utilized by researchers.

Paper	Data Source	Total Spectra	Sample Categories
Shi & Sang (2025)	Created their own	11,147	18 (pure & blends)
Huang et al. (2022)	Commercial fibers	600,404	25 (natural & synthetic)
Kainz et al. (2025)	Created their own	Not specified	12 (pure & blends)

2.1.3 Terahertz (THz) Spectroscopy (0.1–10 THz)

Terahertz Time-Domain Spectroscopy (**THz-TDS**) is a new non-destructive method that works in the space between microwaves and infrared light. THz waves interact with low-frequency intermolecular vibrations, hydrogen-bond networks, and the bulk dielectric properties of materials. This gives us information that we can't get from MIR or NIR.

Principle: THz spectroscopy measures the amplitude and phase of THz pulses that are sent or reflected. This lets you find the absorption coefficients, refractive index, and dielectric constants of textile fibers.

Advantages:

- Less affected by dyes and coloration, unlike NIR where dye absorption distorts spectra.
- Deeper penetration in non-conductive fibers such as cotton, wool, silk, polyester, and nylon.
- Lower scattering compared to MIR when fabrics have rough or textured surfaces.
- Sensitive to structural differences, such as crystallinity and moisture content, which vary across fibre types.

2.2 PRIOR RESEARCH

A lot of research has been done on using spectroscopy to identify textile fibers. This has led to great advancements, but it has also shown that there are still problems with using these techniques in the real world.

Zhou et al. (2019) showed that combining near-infrared (NIR) spectroscopy with traditional chemometric methods like PCA, SIMCA, and LDA works well for sorting common textile fibers.[8] Shi and Sang (2025), in their recent study achieved great F1 scores with ModernTCN across 18 textile classes. They achieved classification accuracy of 93.28% and F1-score of 94.47%. The results obtained by different researchers found out during literature review are shown in the Figure 2. This showed that NIR combined

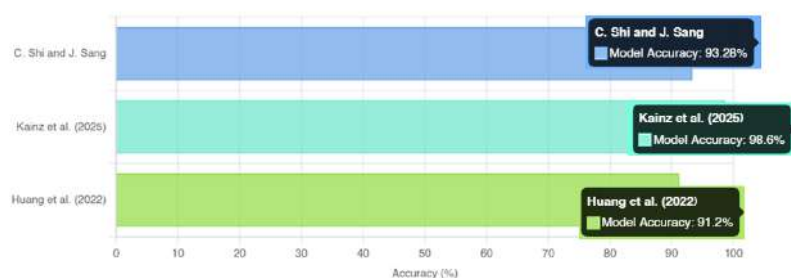


Figure 2: Comparison of classification accuracies reported by different machine learning models in textile research

with machine learning is possibly a rational choice to effectively classify fibers in real time.[4]

Further research by other researchers highlighted the limitations of using NIR. Even with these improvements, NIR still has trouble reliably telling chemically very similar fibers apart (like cashmere and wool or certain cellulosic variants) because their spectra overlap, which often means that targeted bands or specialized models are needed [4][8]

Manley (2014) confirmed the speed of NIR but pointed out that MIR is more accurate. He also talked about problems that can come up with scattering, moisture, and finishing.[3]

Naftaly et al. (2013) [9] showed that THz transmission spectra can tell the difference between wool, cotton, silk, polyester, and acrylic fibers. This means that they can be sorted based on how they respond to electric fields and how much light they absorb. Yoshizumi et al. (2024) [12] demonstrated that THz parameters derived from woollen textiles can effectively characterize fiber density and structure, with diminished interference from dyes, thereby validating THz for practical applications. Recent technology-readiness assessments by Bjarnason et al. (2013) [10] also highlighted THz's potential in analyzing multilayer textiles and composite garments due to its penetrative capability and non-ionizing nature.

MIR spectroscopy (FTIR) is known for being more chemically specific [3]. Peets et al. (2019) conducted significant research on reflectance-FTIR (r-FTIR) for textile analysis, demonstrating its efficacy for fragile heritage items and its superiority over ATR in distinguishing specific amide-containing fibers (e.g., silk, wool, polyamide), identifying a distinct silk peak at approximately 1708cm^{-1} . They emphasized the genuinely non-contact characteristic of r-FTIR while recognizing difficulties associated with extremely coarse or fine fibers [2]. To tackle multi-material complexities, Leitner et al. (2024) examined multi-fabric garments by integrating camera-based segmentation with

point NIR measurements, acknowledging that bulk analysis is inadequate for heterogeneous textiles. For interpreting complex spectra, Enders et al. (2021) demonstrated the efficacy of image-based CNNs in analyzing intricate FTIR data for functional group mapping, indicating the potential of AI in comprehensive textile analysis [7].

2.3 RESEARCH GAPS IDENTIFIED

Although MIR, NIR, and hyperspectral methods have enabled major advances in textile identification, several critical research gaps remain—particularly when considering real-world textile waste and modern garment structures.

- **Difficulty in Identifying Chemically Similar Fibres:** NIR has trouble because of overlapping overtone bands. MIR, on the other hand, is more chemically specific but still has trouble telling the difference between fibers with very similar molecular structures (e.g., cotton vs. viscose, wool vs. cashmere). These materials have similar vibrational properties that often fall within close spectral ranges, which makes it hard to tell them apart.
- **Influence of Dyeing, Finishing, and Surface Conditions:** Dyes on the surface, optical finishes, moisture, and physical texture all have a big effect on both MIR and NIR signals. In real-world recycling settings, fabrics are rarely clean, uniformly colored, or untreated, which makes measured spectra very distorted. This makes MIR/NIR less reliable in places where they aren't controlled.
- **Very Limited Research in Terahertz Spectroscopy for Textiles:** A review of the literature indicates that THz spectroscopy in textiles remains in its nascent phase. There are only a few studies, and most of them look at cotton, wool, polyester, silk, and acrylic. The majority of the work was performed on pristine, homogeneous laboratory specimens. There is very little published THz research on dyed materials, blended fabrics, multilayer garments, or post-consumer waste.
- **Absence of Public THz Textile Datasets:** There are many MIR/NIR datasets, but there are no public THz datasets for textile engineering. Current THz studies comprise merely 5–8 fibers at most, characterized by small sample sizes and restricted spectral diversity. This makes it hard to look into THz classification based on machine learning.
- **Untested Potential of THz for Real-World Sorting:** THz waves go deeper than MIR waves and are less affected by colorants, which makes them theoretically good for sorting. But there haven't been any industrial tests of THz performance on real clothes, like blends, elastics, denim, multi-layer jackets, or printed textiles. This means that there is still a lot of research that needs to be done.

2.4 MOTIVATION

The project started with Mid-Infrared (MIR) imaging because there was MIR equipment available and MIR is known for being very specific to chemicals. Through experimentation, MIR demonstrated its utility in developing an initial baseline classification model. But soon, practical problems arose. The MIR images were very sensitive to the concentration of the dye, the reflectance changed with the texture of the surface, and chemically similar fibers made overlapping absorption patterns that made it harder to classify them.

These observations, along with information from other MIR/NIR studies, showed that MIR by itself might not be strong enough for sorting large amounts of textiles, especially in real-world recycling where fabrics are dyed, worn, blended, and have complicated structures.

This prompted a more thorough examination of **Terahertz (THz) Time-Domain Spectroscopy (THz-TDS)**. THz radiation interacts with textiles in a way that is different from MIR: it goes deeper, is less affected by dyes and colors, and gives access to low-frequency vibrational and dielectric properties that show differences in the bulk structure of fibers. There isn't a lot of THz research on textiles, but the studies that do exist all show that THz signatures are different for cotton, wool, polyester, and acrylic fibers, even when MIR/NIR signals overlap.

These things made us change the direction of the project. MIR experiments are still an important starting point, but the long-term goal of this work is to shift the focus to THz spectroscopy as the main area of research. This project lays the groundwork for creating a more accurate, dye-resistant, and structurally sensitive identification method that can be used for next-generation textile recycling. It does this by looking at the pros and cons of MIR and the possibilities of THz.

Chapter-3

Objectives

Objectives

The overall objective of this project is to investigate the suitability of spectroscopic techniques for textile identification and to transition the research direction toward Terahertz (THz) spectroscopy based on limitations observed in MIR-based methods. The key objectives are as follows:

- **Establish Verified Ground Truth:** Use standardized burn tests and chemical solubility tests to make sure that all labeled samples are chemically accurate. This will help you figure out what the cotton, polyester, and nylon samples are made of.
- **Do Initial MIR Experiments:** Use a 7.5–14 μm camera to take a controlled set of MIR images so you can see how well MIR imaging can tell the difference between different types of textile fibers.
- **Train and Evaluate MIR-Based Models:** Create and test YOLO models trained on MIR image features to measure how accurate they are at classifying data and find out what MIR data is good at and what it isn't.
- **Analyze MIR Limitations in Detail:** Write down how dyes, finishes, texture differences, light effect and chemical similarities affect the accuracy of MIR classification. This will give you a technical basis for moving beyond MIR in the project.
- **Conduct a Comprehensive Review of THz Spectroscopy:** Examine the limited existing THz research on textiles (Naftaly, 2013; Yoshizumi, 2024, etc.) to identify THz absorption characteristics, refractive index patterns, and material-specific behaviors.
- **Define a THz-Centric Experimental Plan for Future Work:** Investigations: Drawing from existing literature and the constraints of MIR, delineate a comprehensive strategy for forthcoming experiments employing Terahertz Time-Domain Spectroscopy as the principal method for textile identification.

Chapter-4

Experimental Section

Experimental Section

4.1 PHASE I: MIR BASED CLASSIFICATION

We chose 3 fabrics for initial study: Cotton, Polyester, and Nylon. To ensure the ground truth for our dataset, each sample was rigorously verified using burn tests and chemical solubility tests.

- We performed two tests for checking:
 - **Burning Test:** We looked at how each sample acted near a flame, such as whether it melted, burned continuously, smelled, and what kind of residue it left behind (e.g., fluffy ash, hard bead).
 - **Chemical Solubility Test:** Acetone (70 percent v/v), 5N HCl, $\text{CaCl}_2 + \text{HCOOH}$, 75 percent Sulphuric Acid, and Concentrated Sulphuric Acid were used to test the samples one after the other. Each fiber exhibited a unique solubility response.

Only materials that showed the expected verification responses were kept; the rest were discarded and not added to the MIR imaging dataset. Table 2 and 3 shows the verification results for Cotton, Nylon, and Polyester for reference.



(a) Polyester showing self-extinguishing behavior.



(b) Suspected nylon sample failing to self-extinguish.



(c) Char formation indicating blended composition.

Figure 3: Burn test results used to verify fabric composition.

Table 2: Burn Test Results for Fabric Verification.

Fibre	Reaction to Flame	Odour	Ash/Residue
Cotton	Burns quickly, continues to glow.	Burning paper	Small, fluffy grey ash.
Nylon	Shrinks, melts, usually self-extinguishing.	Celery-like	Hard grey/brown bead.
Polyester	Shrinks, melts, black smoke, self-extinguishing.	Chemical	Hard, round black bead.

The burn test confirmed the identities by comparing flame behavior, odor, and residue. For chemical verification, the solubility of each fabric was tested in various solvents.

Table 3: Chemical Solubility Test Results.

Solvent	Cotton	Polyester	Nylon
Acetone	Insoluble	Insoluble	Insoluble
5N HCl	Insoluble	Insoluble	Soluble
75% H ₂ SO ₄	Soluble	Insoluble	-
Conc. H ₂ SO ₄	-	Soluble	-

4.1.1 Dataset Creation

An MIR camera operating in the 7.5–14 μm wavelength range was used for image acquisition.



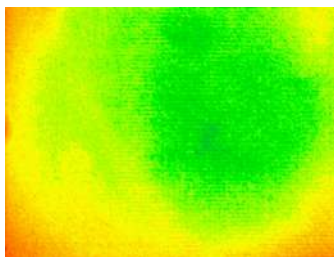
(a) Nylon sample.



(b) Cotton sample.



(c) Polyester sample.



(d) MIR image of Nylon.



(e) MIR image of Cotton.



(f) MIR image of Polyester.

Figure 4: Fabric samples (top row) and their corresponding MIR images (bottom row).

- **Dataset Size:** 120 images total (40 for each fabric type) as of September 20.
- **Image Specifications:** JPG format, with dimensions of 1280x960 pixels.
- **Controlled Conditions:** A fixed camera at a distance of 0.07 meters was maintained, and the ambient temperature was kept within a controlled range of 21–24°C.

4.1.2 Preliminary Modeling Results

After preparing the initial dataset, preliminary classification task was performed using the **YOLO11x-cls** architecture. The model was initialized with pretrained weights to leverage transfer learning. Training was conducted for **100 epochs** with a patience of 100 epochs to prevent overfitting. The input images were resized to a resolution of 224×224 pixels, and a batch size of 16 was utilized. The optimization process employed an initial learning rate (η) of 0.01, a momentum of 0.937, and a weight decay of 0.0005. To improve generalization, data augmentation techniques were applied, including horizontal flipping (50% probability), mosaic augmentation, and random erasing (40% probability).

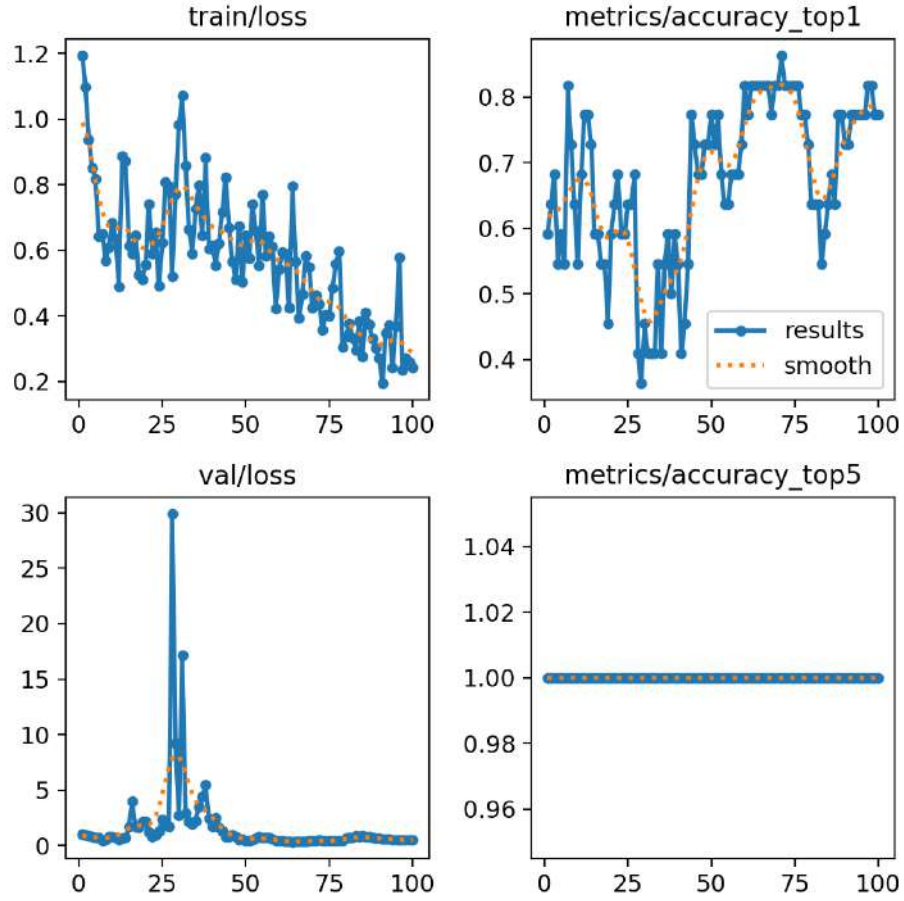


Figure 5: Training performance metrics. The graphs illustrate the training/validation loss and top-1/top-5 accuracy over 100 epochs.

As observed in Figure 5, the model shows convergence with decreasing loss and increasing accuracy. To further evaluate the specific class-wise performance, a normalized confusion matrix was generated (Figure 6).

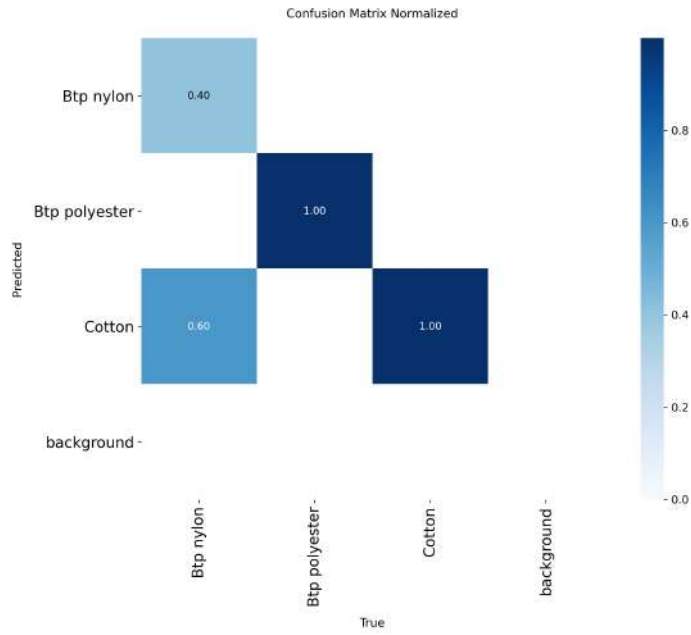


Figure 6: Normalized Confusion Matrix on the validation set. Perfect classification (1.00) is achieved for Polyester and Cotton, while Nylon exhibits some confusion with Cotton in this preliminary iteration.

Finally, a qualitative analysis was performed by visualizing a batch of validation images. Figure 7 presents a side-by-side comparison of the Ground Truth labels and the Model Predictions, demonstrating the model’s ability to spatially recognize and classify the texture patterns of the fabric samples.

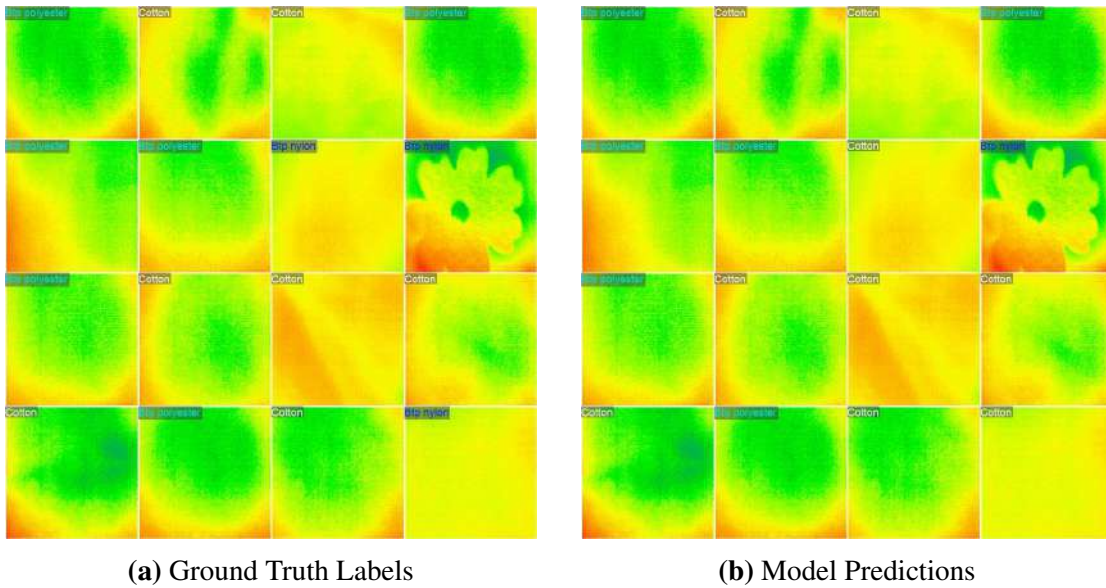


Figure 7: Comparison of Ground Truth labels versus Model Predictions.

4.1.3 Limitations of Passive MIR and Pivot to Terahertz Spectroscopy

The first MIR models worked well during training, but when they were tested on random, uncontrolled samples, they weren't very reliable. When the environment changed, the classification wasn't as accurate. The hypothesis asserted that reliance on passive MIR imaging represented the primary factor, considering that natural ambient infrared radiation often does not yield a high signal-to-noise ratio. To solve this problem, we first suggested using Active MIR Imaging with a strong, single-frequency light source to block out background radiation. We talked to **Prof. Amartya Sengupta** (Department of Physics, IIT Delhi), who is an expert in imaging spectroscopy, because it was hard to get the hardware needed for such a source. He said that MIR is chemically specific, but surface dyes and changes in temperature can still change it. He said that **Terahertz (THz) Spectroscopy** would be a better choice because the THz range (0.1–10 THz) isn't very affected by fabric dyes and is less likely to be affected by optical noise.

4.2 PHASE II: TERAHERTZ (THZ) SPECTROSCOPY CHARACTERIZATION

After shifting to Terahertz spectroscopy, a comprehensive experimental set was designed to validate its effectiveness, particularly concerning dye independence.

4.2.1 Sample Preparation

A total of 6 cotton-based samples were prepared for THz analysis to represent various real-world conditions:

1. **Laboratory-Dyed Samples:** Three standard cotton samples were individually dyed in the laboratory using three different classes of dyes: **Direct dye, Vat dye, and Reactive dye.**
2. **Commercial Samples:** Three samples were procured from the market: standard White Cotton, commercially dyed Pink Cotton, and a Cotton-Viscose blend.

4.2.2 Experimental Setup and Results

The prepared samples were subjected to Terahertz spectroscopy. To ensure a high signal-to-noise ratio and robust spectral data, the measurements were conducted with **8000 averages** per sample scan.

Figure 8 presents the resulting absorbance spectra for all six samples across the frequency range of approximately 0.5 THz to 7.5 THz. The plot illustrates the spectral response of the differently dyed and blended cotton fibers in the Terahertz regime.

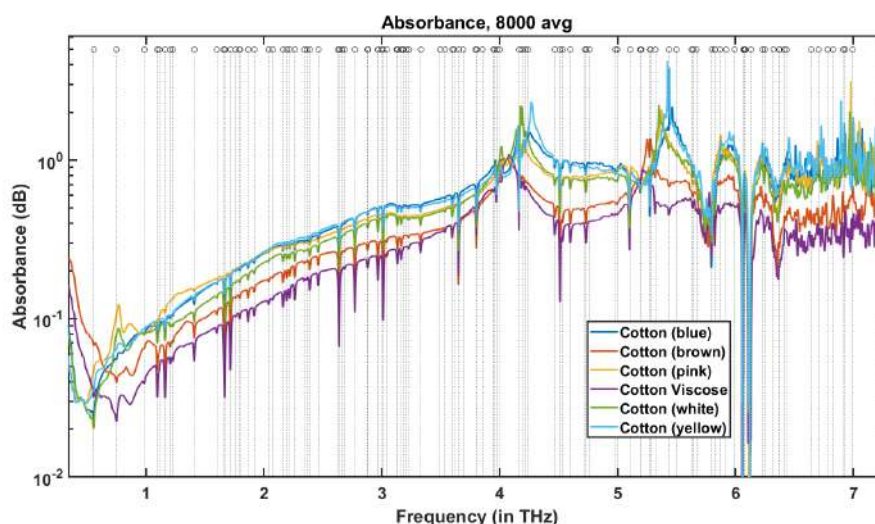


Figure 8: Absorbance spectra (in dB) of dyed and blended cotton samples obtained via Terahertz spectroscopy with 8000 averages. The legend indicates the specific dye or blend for each curve.

Initial observation of the spectra in Figure 8 allows for the analysis of common underlying features attributable to the cotton base, alongside variations introduced by the different dyes and the viscose blend.

4.2.3 Comparative Analysis of Fiber Classes

Following the dye-independence study, the experimental emphasis transitioned to the principal aim: the differentiation of the three fundamental material categories—**Cotton** (cellulosic), **Nylon** (polyamide), and **Polyester** (synthetic polymer).

The previously validated samples of pure Cotton, Nylon, and Polyester were examined under uniform Terahertz spectroscopic conditions. The goal was to find unique spectral signatures and absorption peaks that don't change no matter what form the sample takes.

Figure 9 shows how the three materials' absorbance spectra compare to each other. The analysis shows that each type of fiber has a different spectral behavior. Nylon and polyester have specific absorption peaks at certain frequencies that make them easy to tell apart from cotton and from each other.

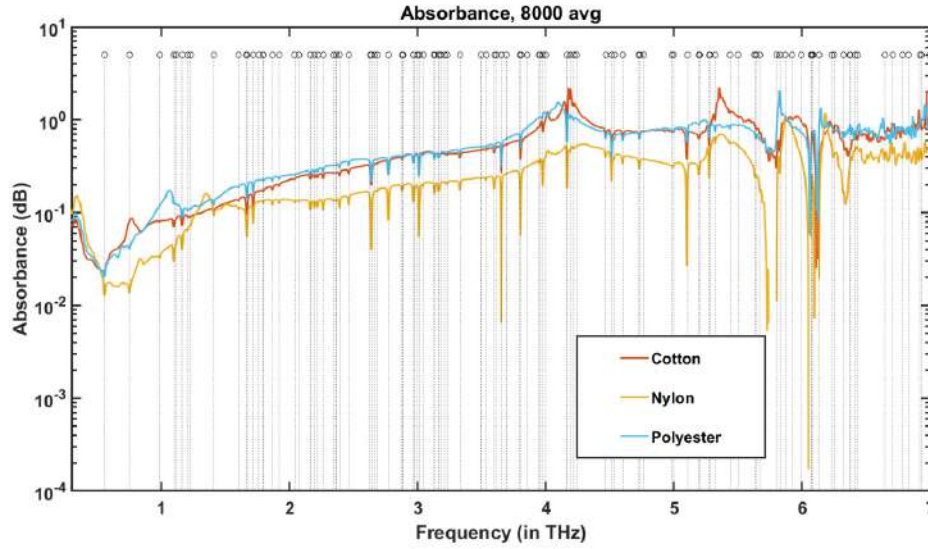


Figure 9: Comparative Terahertz absorbance spectra of Cotton, Nylon, and Polyester. The arrows/highlights indicate the distinct absorption peaks identified for the discrimination of synthetic fibers.

The specific absorption peaks identified for each material class are summarized in Table 4. These distinct frequency markers serve as the quantitative basis for distinguishing the fibers.

Table 4: Identified Terahertz Absorption Peaks (in THz) for Fiber Samples.

Polyester	Nylon	Viscose	Cotton
1.068	1.346	4.500	0.641
1.254	1.543	0.737	4.238
1.690	4.309	—	0.480

These results confirm that Terahertz spectroscopy provides sufficient spectral resolution to distinguish between these chemically diverse fibers based on their structural properties rather than surface appearance.

Chapter-5

Summary and Conclusions

Summary and Conclusions

This report shows how the project on non-destructive textile identification using spectroscopic methods and machine learning is doing so far. The project started with a thorough study of MIR, NIR, and THz methods, and then moved on to experiments with Mid-Infrared (MIR) imaging. We made sure that samples of cotton, polyester, and nylon were real, and we made a controlled MIR dataset with 120 images. A YOLO11xcls model that was trained on this dataset did well in controlled tests.

However, MIR imaging showed clear problems when it was tested on randomly captured samples. Because it was sensitive to background thermal noise, dye concentration, and fabric texture, its predictions were not always accurate. These problems, along with evidence from the literature, showed that MIR by itself might not be strong enough for sorting on a large scale.

To solve this problem, the focus of the research changed to Terahertz (THz) spectroscopy. THz tests were done on dyed cotton samples and on cotton, nylon, and polyester. The results showed different absorption patterns and less effect of dyes. The findings indicated that THz radiation can distinguish textile fibers based on more profound structural and dielectric characteristics.

In conclusion, although MIR served as a valuable reference point, THz spectroscopy proved to be a more advantageous method for accurate textile identification. The next step in the project will be to improve THz measurements, find unique spectral features, and make THz-based classification models that can be used in the future for recycling textiles.

References

- [1] Fashion for Good, Sorting for Circularity: India – Wealth in Waste. Complete URL: <https://www.fashionforgood.com/report/wealth-in-waste/>[Accessed on August 22, 2025].
- [2] Peets, P., Kaupmees, K., Vahur, S., Leito, I., Reflectance FT-IR spectroscopy as a viable option for textile fiber identification. *Heritage Science*, 2019, 7, 93.
- [3] Manley, M., Near-infrared spectroscopy and hyperspectral imaging: non-destructive analysis of biological materials. *Chemical Society Reviews*, 2014, 43, 8200–8214.
- [4] Shi, C., Sang, J., Qualitative recognition of waste textiles based on near-infrared spectroscopy and ModernTCN. *Microchemical Journal*, 2025, 214, 113902.
- [5] Kainz, M., Krondorfer, J.K., Jaschik, M., Jernej, M., Ganster, H., Supervised and unsupervised textile classification via near-infrared hyperspectral imaging and deep learning. *arXiv preprint*, arXiv:2505.03575, 2025.
- [6] Huang, J., Li, Y., Gao, S., et al., Non-destructive detection and classification of textile fibres based on hyperspectral imaging and 1D-CNN. *Analytica Chimica Acta*, 2022, 1224, 340238.
- [7] Enders, A.A., North, N.M., Fensore, C.M., Velez-Alvarez, J., Allen, H.C., Functional group identification for FTIR spectra using image-based machine learning models. *Analytical Chemistry*, 2021, 93, 9711–9718.
- [8] Zhou, J., Yu, L., Ding, Q., Wang, R., Textile fiber identification using near-infrared spectroscopy and pattern recognition. *AUTEX Research Journal*, 2019, 19(2), 201–209.
- [9] Naftaly, M., Molloy, J.F., Lanskii, G.V., Kokh, K.A., & Andreev, Yu.M. Terahertz time-domain spectroscopy for textile identification. *Applied Optics*, 52, 4433–4436, 2013.
- [10] Bjarnason, J.E., Chan, T.L.J., Lee, A.W.M., Celis, M.A., & Brown, E.R. Millimeter-wave, terahertz, and mid-infrared transmission through common clothing. *Applied Physics Letters*, 85, 519–521, 2004.

- [11] Notario, A., Hernández-Andrés, J., Romero, J., García-Heras, J.L., Gómez-Robledo, L., & López-Rodríguez, A. Hyperspectral imaging for textile recycling in the visible and near-infrared ranges. *Journal of Spectral Imaging*, 8, Article A08, 2019.
- [12] Yoshizumi, T., Iwasaki, K., Fujii, S., Kimura, T., Yamamoto, M., Manago, G., Yu, J., & Tanabe, T. Extraction of terahertz wave parameters that characterize woollen clothes. *Textile Research Journal*, 95, 399–403, 2024.
- [13] Yan, C., Yang, B., & Yu, Z. Terahertz time-domain spectroscopy for the identification of two cellulosic fibers with similar chemical composition. *Analytical Letters*, 46, 946–958, 2013.
- [14] Kundu, S., Bera, K., Ghosh, S., & Dhar, A. Terahertz waves and their applications in textiles and composites. *Advanced Materials Technologies*, 10, (pages vary), 2025.